Advanced information, computation, communication I
EPFL - Fall semester 2026-2027

Week 9

November 13, 2026

Exercise 1. Use strong induction to show the following statement:
Any postage of at least 8 cents can be formed using just 3 cents and 5 cents stamps.

Exercise 2. Denote by f_n the nth Fibonacci number, i.e., $f_0 = 0, f_1 = 1$ and for $n \geq 2$, $f_n = f_{n-1} + f_{n-2}$. Prove that $f_1^2 + f_2^2 + \cdots + f_n^2 = f_n f_{n+1}$, when n is a positive integer.

Exercise 3. Find the mistake in the following proof:
Let $P(n)$ for $n \in \mathbf{Z}_{\geq 0}$ be the propositional function "all sets composed of n integers consist of only even integers," which is proved using strong induction:

Basis step $P(0)$ is true, since if S is an empty set of integers the statement "$\forall s\ s \in S \rightarrow$ s is even" is true.

Inductive step Let $k \geq 0$ and assume that $P(i)$ is true for $0 \leq i \leq k$. To prove that $P(k+1)$ is true we use the following steps:

1. Let T be an arbitrary set of integers with $|T| = k + 1$.
2. Write T as the disjoint union of sets T_1 and T_2 such that $|T_1| = k$ and $|T_2| = 1$.
3. Because $|T_1| < |T|$ and $|T_2| < |T|$ the induction hypothesis applies to both T_1 and T_2, implying that all elements of both T_1 and T_2 are even.
4. Because $T = T_1 \cup T_2$ it follows that all elements of T are even as well.
5. Because T was arbitrarily chosen as a set of integers of cardinality $k + 1$, it follows that $P(k + 1)$ is true.

Because not all integers are even, the proof cannot be correct.

Exercise 4. Let $P(n)$ for $n \in \mathbf{Z}_{>0}$ be the propositional function "all sets composed of n integers consist of only odd integers," which is proved using strong induction:

Basis Step $P(1)$ is true because 1 is odd.

Inductive step Let $k > 0$ and assume that $P(i)$ is true for $0 < i \leq k$. To prove that $P(k+1)$ is true we use the following steps:

1. Let S be an arbitrary set of integers with $|S| = k + 1$.
2. Write S as the disjoint union of sets S_1 and S_2 such that $|S_1| = k$ and $|S_2| = 1$.
3. Because $|S_1| < |S|$ and $|S_2| < |S|$ the induction hypothesis applies to both S_1 and S_2,
4. implying that all elements of both S_1 and S_2 are odd.
5. Because $S = S_1 \cup S_2$ it follows that all elements of S are odd as well.
6. Because S is an arbitrarily chosen set of integers with $|S| = k + 1$, it follows that $P(k + 1)$ is true.

◯ Only the basis step in the proof is incorrect.

◯ The basis step and step (3) of the inductive step of the proof are incorrect.

◯ Only step (3) of the inductive step of the proof is incorrect.

◯ Only step (4) of the inductive step of the proof is incorrect.

Exercise 5. The **Lucas numbers** satisfy the recurrence relation

$$L_n = L_{n-1} + L_{n-2}$$

and the initial conditions $L_0 = 2$ and $L_1 = 1$. Show that $L_n = f_{n-1} + f_{n+1}$ for $n = 2, 3, ...$, where f_n is the nth Fibonacci number.

Exercise 6. **[Exam 2022]** Let us prove that each n distinct points on the plane lie on the same line.

Basis Step Any 1 or 2 distinct points lie on the same line (axioms of the geometry).

Inductive step Let $k > 1$ and assume that the statement is true for $0 < i \leq k$. To prove the statement for $k + 1$, we use the following steps:

1. Let $A_1, A_2, \ldots, A_{k+1}$ be arbitrary $k + 1$ points.
2. By the induction hypothesis, points $A_1, A_2, \ldots, A_k$ lie on the same line l.
3. By the induction hypothesis, points $A_2, A_3, \ldots, A_{k+1}$ lie on the same line l'.
4. From the axioms of geometry, only one line can pass through two distinct points on the plane. As A_2 and A_k both lie on the lines l and l', we obtain that lines l and l' are the same.
5. Therefore, all the points $A_1, \ldots, A_{k+1}$ lie on the same line, and the induction hypothesis is true for $k + 1$ points.

What is wrong with the "proof"?

Exercise 7. Let n be a positive integer and x be an integer.

1. Give a recursive algorithm for computing nx (you can only use addition).
2. Prove that your recursive algorithm is correct.

Exercise 8. The reversal of a string is the string consisting of the symbols of the string in reverse order. The reversal of the string w is denoted by w^R.

1. Find the reversal of the following bit strings
   (a) 0101
   (b) 1 1011
   (c) 1 0000 1001 0111
2. Give a recursive definition of the reversal of a string.
3. Use structural induction to prove that $(w_1 w_2)^R = w_2^R w_1^R$.

Exercise 9. Let $\{a_n\}$ be the sequence defined by $a_0 = 1$, $a_1 = 2$, and $a_n = a_{n-1} \times a_{n-2}$, for n = 2,3,4,$\cdots$

1. Describe an iterative algorithm for finding the nth term of $\{a_n\}$
2. Describe a recursive algorithm for finding the nth term of $\{a_n\}$

Exercise 10. **[Exam 2023]** Let set S be a set of strings made by two letters a and b. The set is recursively defined as follows:

Basis step: $b \in S, baa \in S$.

Recursion step: $s \in S \rightarrow \ (bbs \in S) \wedge (aas \in S)$

Prove by structural induction that every element in S has an even number of a and an odd number of b.